LendingClub Loan Default and Profitability Prediction

Team 14 | UE24CS352A - Machine Learning | Problem No. 39
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GitHub: https://github.com/karan21v/LendingClub-P2P-Loan-Default-Prediction

1. Problem Statement

Peer-to-peer (P2P) lending investors must judge both repayment risk and potential return before funding a loan. We use historical LendingClub data to build (i) a binary classifier for Fully Paid versus Default/Charged Off and (ii) a regression model for Net Annualized Return (NAR). The two outputs support a simple loan-ranking strategy: lower default risk indicates credit quality, while predicted NAR provides an investment-oriented ranking.

2. Dataset Details

The accepted LendingClub archive is read in chunks, but only finalized loans issued during 2012-2015 are retained. Current and late loans are excluded because their final outcome is unknown. Post-loan variables such as total_pymnt and last_pymnt_d are reserved only for constructing historical NAR and are not used as predictors.

Property	Observed value
Finalized loans	786,820 (2012: 53,367; 2013: 134,804; 2014: 223,103; 2015: 375,546)
Class balance	640,186 Fully Paid (81.36%); 146,634 Default/Charged Off (18.64%)
Loaded data	33 relevant columns; about 198 MB after filtering
Model inputs	21 numeric + 8 categorical features after engineering; one-hot encoding expands categoricals
Main missingness	emp_length about 5.3%, mort_acc about 1%; remaining numeric gaps are small

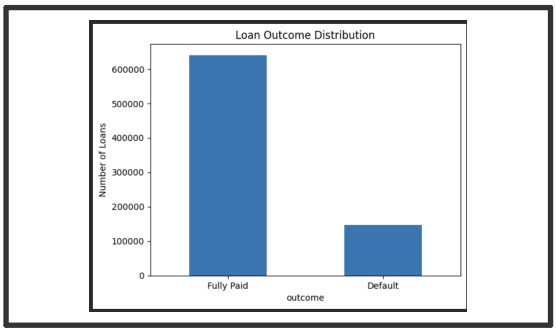


Fig. 1. Finalized-loan outcome distribution shows the 81/19 class imbalance.

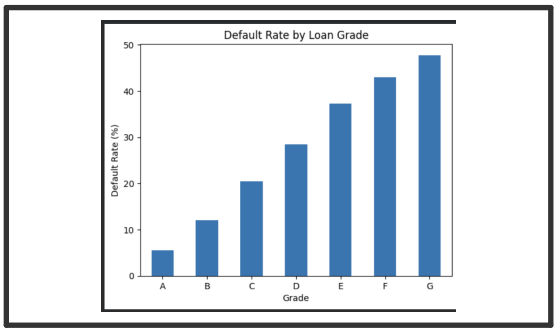


Fig. 2. Default rate rises strongly from grade A to grade G, validating risk signal in the data.

3. Approach

- **Feature engineering.** term is converted to months; employment length to years; FICO low/high to a midpoint score; credit-history age is derived from issue date and earliest credit line; free-text emp_title is dropped.
- **Preprocessing.** numeric features use median imputation and scaling; categorical features use most-frequent imputation and one-hot encoding inside scikit-learn pipelines.
- **Split and compute.** 70% train / 15% validation / 15% test. For Colab feasibility, model fitting uses a stratified 200,000-loan sample from the training partition; validation and test partitions remain separate.
- **Classification.** Dummy baseline, Logistic Regression with balanced class weights, MLP, and Random Forest are compared. Because defaults are the minority, model selection emphasizes default-class F1 rather than accuracy alone.
- **Regression.** NAR is derived from loan amount, total payment and loan duration. Linear, Ridge, MLP, and Random Forest regressors (depth 4/8/10) are compared by validation R2, followed by a loan-ranking simulation.

4. Implementation and Experimental Results

Classification (test)	Value	Regression / strategy	Observed result
Selected classifier	Logistic Regression	Best validation regressor	Neural Network
Default precision / recall	0.327 / 0.589	Best validation R^2	≈0.040
Default F1	0.421	Investment threshold	7.16% predicted NAR
Accuracy / weighted F1	0.698 / 0.726	Ranking behaviour	Top-ranked subsets show higher realized NAR

Classification interpretation. On the 118,023-loan held-out test set, Logistic Regression correctly detected 12,948 defaults and missed 9,047; 26,627 fully paid loans were falsely flagged. The operating point therefore favors recall over precision, making the model useful for screening risky loans.

Profitability and GUI. Exact NAR prediction is difficult (validation R^2 around 0.04), but ranking remains useful: the top-ranked loan subsets achieved higher realized historical NAR. The Gradio interface loads the trained models and accepts custom borrower/loan inputs to display default probability, repayment prediction, predicted NAR and INVEST / DO NOT INVEST.

5. Conclusion and Limitations

Accuracy alone is misleading on the 81/19 data; default-class recall and F1 are necessary for evaluation. Logistic Regression gives the strongest default-class F1 in our tested setup. NAR is more useful as a ranking signal than as an exact forecast. Main limitations are the historical 2012-2015 window, the 200,000-row fitting sample, and use of LendingClub-assigned grade information. **Reference:** [1] P. Li and G. Han, “LendingClub Loan Default and Profitability Prediction,” Stanford University (supplied reference paper).